A Statistical Analysis of Vehicle Data

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Tools Used: R, RStudio, LaTeX

Executive Summary

This project uses linear regression analysis to identify which vehicle characteristics are associated with vehicle prices. After exploring the data, multiple linear regression models were built and compared using cross-validation, model selection, and diagnostic checks. The final model provides interpretable insights that can help support pricing decisions, market analysis, and valuation strategy. The final model identified weight, horsepower, drive wheels, vehicle type, and the horsepower-by-weight interaction as important factors associated with MSRP.

1 Business Objective and Background

In today's car market, understanding the factors associated with vehicle prices can help buyers make more informed decisions when purchasing a vehicle, select vehicles for business use, and help dealerships and manufacturers evaluate market pricing patterns. This project uses multiple linear regression to identify which vehicle characteristics are associated with MSRP and to estimate the strength of those relationships. These insights can support business owners, dealerships, and customers in making data-informed pricing and purchasing decisions.

2 Data Overview

The original dataset contained 429 observations and 14 variables. It includes two potential response variables: MSRP, which stands for Manufacturer's Suggested Retail Price, and dealer cost. MSRP was selected as the main response variable because it represents the listed retail price of the vehicle and provides a clear measure of car price for the analysis.

Predictor variables included engine size, number of cylinders, horsepower, vehicle type, fuel type, and other vehicle characteristics. These variables were used to examine which features are most strongly associated with differences in MSRP.

3 Data Cleaning and Exploratory Data Analysis

Before modeling, the dataset was cleaned and prepared to make the variables consistent, usable, and appropriate for regression analysis. This included standardizing variable names, converting categorical variables into factors, reviewing missing or abnormal values, and grouping sparse categories that did not provide enough information for reliable model estimates. Since MSRP was selected as the main response variable, the remaining vehicle characteristics were prepared as predictors for the regression analysis.

After preprocessing, the data were explored to understand pricing patterns before modeling. This included reviewing the distribution of MSRP, checking for unusual values, and comparing price patterns across key vehicle characteristics such as horsepower, engine size, and vehicle type. These checks helped identify skewness, outliers, and important relationships to account for in the regression models. The most important EDA visuals and interpretations are provided in Appendix A.

4 Modeling Approach

After cleaning and exploring the data, linear regression-based modeling was used to estimate how vehicle characteristics were associated with MSRP. Multiple candidate models were compared to evaluate the interpretability of results and predictive performance. Before fitting the model, the data were split into a training set and a testing set. Due to the sample size, cross-validation was performed within the training set to support model selection. Then the final model selected was evaluated on the unseen test set.

Candidate models included a baseline multiple linear regression model, an updated model with a log-transformed response variable, model selection variants with interaction terms, and ridge regression.

Model performance was evaluated using root mean squared error (RMSE), mean absolute error (MAE), and R-squared (R^2). Because the main objective was to assess which vehicle characteristics were associated with vehicle pricing, model assumptions and diagnostic checks were reviewed to support a valid interpretation. Definitions and formulas for the evaluation metrics are provided in Appendix D . Interpretability was considered alongside predictive performance to support appropriate recommendations.

Additional technical details, including the full modeling workflow and documented code, are available in the project repository.

5 Model Results

After comparing the candidate models, the final model was selected based on prediction accuracy, interpretability, and diagnostic performance. The selected model was a multiple linear regression model with interaction terms, which provided a reasonable balance between explaining MSRP differences and producing results that could be interpreted in a business context. A summary of the candidate model comparison is provided in Appendix B, and the final model diagnostic checks are provided in Appendix C.

The most influential predictors in the final model included weight, horsepower, drive wheels, vehicle type, and the horsepower-by-weight interaction. These results suggest that MSRP is associated with a combination of vehicle size, performance, and vehicle-category characteristics rather than one single vehicle feature. Additional coefficient details are provided in Table 2 in the appendix.

Overall, the final log-scale model achieved an $R^2 \approx 0.8562$, an RMSE $\approx$ 0.19598, and an MAE $\approx$ 0.1336 on the test set. Compared with the cross-validation results, the test set performance suggests that the model generalized reasonably well to unseen data.

The MAE of 0.1336 on the log scale corresponds to an approximate average absolute percentage error of 14.3%, while the RMSE of 0.1960 corresponds to an approximate error of 21.7%. For example, for a vehicle priced around $30,000, these percentages correspond to about $4,290 and $6,510, respectively.

More details about the model metrics used, such as definitions and formulas, can be found in the Appendix D

6 Key Findings

The analysis found that vehicle pricing was associated with a combination of size, performance, and vehicle-category characteristics.

- Weight, horsepower, drive wheels, and vehicle type were among the strongest predictors associated with MSRP.
- The horsepower-by-weight interaction suggested that the relationship between performance and price may vary depending on vehicle weight.
- MSRP should not be evaluated using one feature alone; a more reliable pricing assessment should consider multiple vehicle characteristics at the same time.

7 Business Recommendations

Based on the model results, vehicle pricing decisions should consider multiple characteristics together rather than relying on a single feature. Weight, horsepower, drive wheels, and vehicle type were important factors associated with MSRP, so these variables should be reviewed closely when evaluating vehicle prices.

For dealerships, manufacturers, or pricing analysts, the model can be used as a supporting tool to identify whether a vehicle's listed price is consistent with its characteristics. However, the model should be used alongside market knowledge, vehicle condition, brand reputation, and current demand.

To improve future pricing accuracy, additional observations and more variables should be collected. Future work could also compare the linear regression model with nonlinear approaches, such as machine learning methods and neural networks, to determine whether predictive performance improves.

8 Limitations

This analysis has several limitations. First, the results show associations between vehicle characteristics and MSRP, but they should not be interpreted as direct causal effects. For example, horsepower, weight, and vehicle type may be related to MSRP, but the model does not prove that changing one feature alone would directly change the vehicle price.

Second, MSRP represents the listed manufacturer price, not necessarily the actual transaction price paid by customers. Market conditions, dealer incentives, vehicle condition, brand reputation, and regional demand may also affect real-world pricing but were not fully captured in the dataset.

Third, although diagnostic checks were used to evaluate the final model, regression models rely on assumptions such as linearity, constant error variance, and approximately normal residuals. Any violations of these assumptions may affect the reliability of inference. Diagnostic checks did not indicate severe assumption violations, but the results should still be interpreted with caution.

Finally, the model was built using the available variables in the dataset, so omitted factors may limit its predictive accuracy. The results should therefore be used as a supporting tool for pricing decisions rather than as the sole basis for final valuation.

9 Conclusion

This analysis found that MSRP is associated with multiple vehicle characteristics, including weight, horsepower, drive wheels, and vehicle type. The final model provided a reasonable balance between predictive performance and interpretability, making it useful as a supporting tool for pricing and valuation decisions. However, the results should be used alongside market knowledge and other business factors not included in the dataset.

10 Appendix

A Additional Exploratory Data Analysis

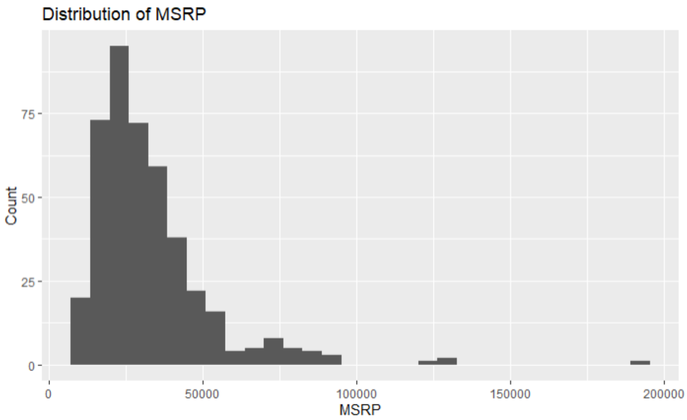

Figure 1: Distribution of MSRP

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Figure 1 shows the distribution of MSRP before modeling. The distribution is right-skewed, indicating that most vehicles are concentrated at lower price levels, while a smaller number of vehicles have much higher listed prices. The small number of higher-priced vehicles may have a strong influence on model estimates.

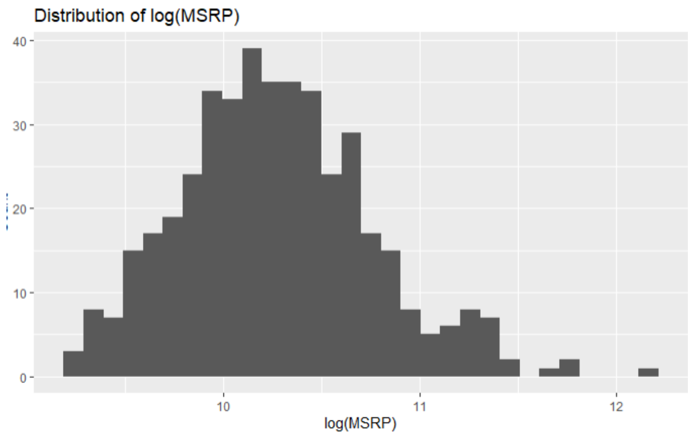

Figure 2: Distribution of log MSRP

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Figure 2 shows the distribution of MSRP after applying a log transformation. Compared with the original MSRP distribution, the log-transformed values appear more balanced and less right-skewed. This supports the use of a log-transformed response variable in some of the candidate regression models.

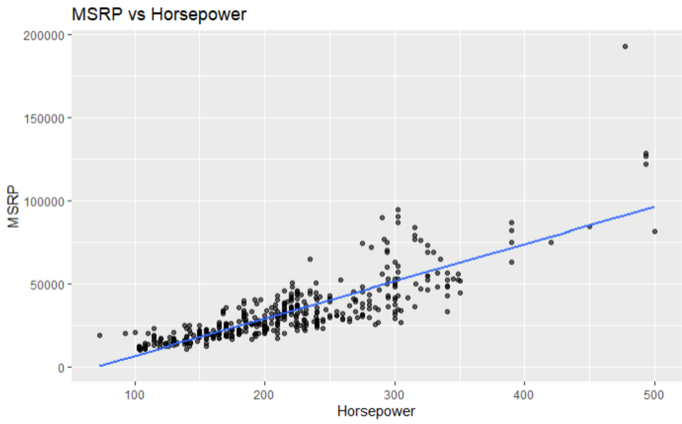

Figure 3: Scatterplot of MSRP and Horsepower

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Figure 3 shows the relationship between horsepower and MSRP. A positive trend is visible, suggesting that vehicles with higher horsepower generally tend to have higher listed prices. However, the wider spread of MSRP values at higher horsepower levels indicates that horsepower alone does not fully explain vehicle pricing. This supports the need to consider horsepower alongside other vehicle characteristics in the regression model.

B Model selection results

Table 1: Candidate model performance comparison

Model	RMSE	MAE	R-squared
Interaction LM	9642.46	6315.02	0.72
Ridge Main Effects	9793.36	6375.11	0.72
Ridge Interaction	9886.17	6417.47	0.71
Main Effects LM	10083.10	6630.57	0.72
Updated LM	10518.82	7005.50	0.70

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Table 1 shows the model selection results. The models were sorted by the MAE. The interaction linear model had the lowest MAE among the candidate models, with an MAE of 6315.02. It is important to note that the model selection was measured by using the MSRP and not the log-transformed version.

C Final model results and diagnostic checks

Table 2 presents the coefficient estimates from the interaction linear regression model using log(MSRP) as the response variable. The estimate column shows the direction and size of the predictor associated with the log price, while the p-value column indicates whether the predictor was statistically significant, hold-

Predictor	Estimate	Std. Error	t-value	p-value	Sig.
`(Intercept)`	10.14606	0.03406	297.908	$< 2e-16$	***
`Engine_Size`	-0.15233	0.04058	-3.754	0.000209	***
`Horsepower`	0.30202	0.03758	8.036	$2.16e-14$	***
`Weight`	0.36006	0.03655	9.851	$< 2e-16$	***
`Width`	-0.08080	0.02695	-2.998	0.002942	**
`out_Horsepower`	0.26251	0.12156	2.160	0.031602	*
`out_Length`	0.16608	0.07941	2.091	0.037333	*
`na_ind_Length`	0.24183	0.18836	1.284	0.200187	
`na_ind_Width`	-0.44058	0.14494	-3.040	0.002577	**
`na_ind_Combined_MPG`	-0.12470	0.07415	-1.682	0.093692	.
`Number_of_Cylinders_X8`	0.15351	0.04234	3.626	0.000339	***
`Drive_Wheels_4WD`	0.07662	0.03383	2.265	0.024233	*
`Drive_Wheels_RWD`	0.14281	0.03308	4.317	$2.15e-05$	***
`Vehicle_Type_Pickup`	-0.30842	0.06990	-4.826	$2.22e-06$	***
`Vehicle_Type_Sedan`	0.09294	0.03672	2.531	0.011889	*
`Vehicle_Type_Sports.Car`	0.29539	0.05979	4.940	$1.30e-06$	***
`Vehicle_Type_SUV`	-0.09978	0.04697	-2.125	0.034446	*
`Horsepower_X_Vehicle_Type_Sedan`	0.08067	0.03174	2.542	0.011530	*
`Length_X_Width`	-0.05072	0.01825	-2.779	0.005798	**
`Horsepower_X_Weight`	-0.07921	0.02213	-3.580	0.000401	***
`Length`	-0.02232	0.02591	-0.861	0.389701	

Table 2: Final Model Inferential Test Results

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ing the other variables constant. Positive estimates are associated with higher MSRP, while negative estimates are associated with lower MSRP, holding other predictors constant.

The strongest significant predictors included Weight, Horsepower, drive wheels, vehicle type, and horsepower-by-weight interaction. These results suggest that MSRP is strongly associated with vehicle size, vehicle category, and performance-related characteristics.

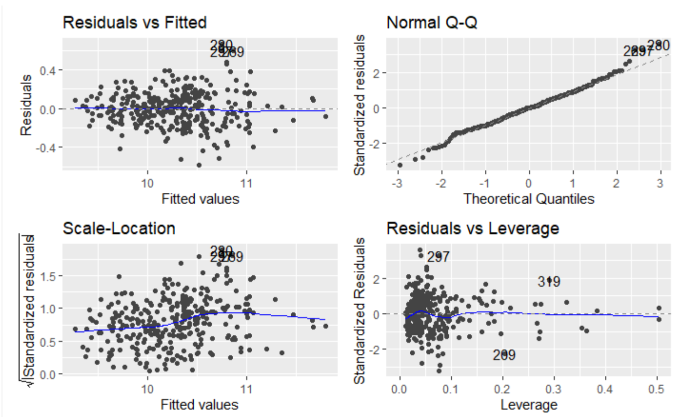

Figure 4: final model diagnostic checks

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Figure 4 shows the final model assumption checks. Based on the residuals-versus-fitted plot, the residuals appear to be randomly scattered around zero without a strong pattern, suggesting that the linearity and constant variance assumptions are reasonably acceptable. The Normal Q-Q plot shows that most points fall close to the reference line, although some deviation appears in the tails. Overall, the diagnostic plots suggest that the final model is appropriate for interpretation, with some caution due to minor departures from normality.

D Metrics used for evaluation

- Root mean squared error (**RMSE**) - A metric that is used to measure the model's average error by penalizing the larger errors.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- Mean absolute error (**MAE**) - A metric that is used to measure the model's average error, using absolute error rather than squared error.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- R-squared (R^2)- measures the proportion of the variation in the outcome variable that is explained by the model.

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y}_i)^2}$$

- Approximate percentage error - Gives the log-scale MAE or RMSE percentage error, which tells the approximate error size on the original scale.

$$\text{Approximate percentage error} = \left(e^{\text{log-scale error}} - 1\right) \times 100$$

Note: This also works for the coefficient of the model as well.

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